BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Keyword 1, Keyword 2, Keyword 3
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1 Introduction

2 Methodology

Describe your structural VAR (SVAR) models and the Monte Carlo Data Generating Processes (DGPs) here.

2.1 Data Generating Process

Explain the true DGP based on Johansen and Juselius (1990), Ramey (2011), or Kilian and Murphy (2014).

3 Simulation Results

Present the results of your Monte Carlo simulations.

Table 1: Relative Mean Squared Error: ICs vs. BVARs.

True DGP (p0)	Estimator Sample Size (T)	AIC	SIC (BIC)	HQC	BVAR (Minn.)	BVAR (Conj.)
4	80	5.700	1.235	5.372	4.974	5.118
	100	1.641	1.020	1.034	2.137	2.178
	120	1.117	1.105	1.018	1.844	1.857
	160	1.096	1.159	1.085	1.593	1.709
	200	1.119	1.170	1.141	1.526	1.621
6	80	3.874	0.996	3.691	4.576	4.549
	100	1.570	1.038	0.997	2.271	2.281
	120	0.960	1.043	0.933	1.684	1.791
	160	0.949	1.005	0.982	1.469	1.564
	200	1.028	1.107	1.071	1.513	1.433
8	80	2.735	0.816	2.567	3.528	3.617
	100	1.264	0.848	0.820	1.706	1.708
	120	0.919	0.965	0.838	1.424	1.459
	160	0.899	0.916	0.895	1.415	1.326
	200	0.926	0.897	0.855	1.299	1.216
10	80	2.302	0.605	2.214	2.501	2.469
	100	1.228	0.714	0.686	1.503	1.500
	120	0.829	0.715	0.699	1.184	1.109
	160	0.802	0.847	0.831	1.211	1.200
	200	0.822	0.761	0.766	1.009	1.065

Table 2: Relative Mean Squared Error: ICs vs. BMA.

True DGP (p0)	Estimator Sample Size (T)	AIC	SIC (BIC)	HQC	BMA (BIC-Weighted)
4	80	5.700	1.235	5.372	0.897
	100	1.641	1.020	1.034	0.972
	120	1.117	1.105	1.018	1.040
	160	1.096	1.159	1.085	1.031
	200	1.119	1.170	1.141	1.120
6	80	3.874	0.996	3.691	0.795
	100	1.570	1.038	0.997	1.013
	120	0.960	1.043	0.933	1.008
	160	0.949	1.005	0.982	0.928
	200	1.028	1.107	1.071	1.009
8	80	2.735	0.816	2.567	0.628
	100	1.264	0.848	0.820	0.769
	120	0.919	0.965	0.838	0.795
	160	0.899	0.916	0.895	0.798
	200	0.926	0.897	0.855	0.788
10	80	2.302	0.605	2.214	0.479
	100	1.228	0.714	0.686	0.663
	120	0.829	0.715	0.699	0.713
	160	0.802	0.847	0.831	0.822
	200	0.822	0.761	0.766	0.648

4 Conclusion

Summarize the main findings and implications for applied macroeconomic research.